

4	(a) electric field strength = force / positive charge	B1	[1]
	(b) (i) at least three equally spaced parallel vertical lines direction down	B1 B1	[2]
	(ii) $E = 1500 / 20 \times 10^{-3} = 75000 \text{ V m}^{-1}$	A1	[1]
	(iii) $F = qE$ ($W = mg$ and) $qE = mg$ $q = mg / E = 5 \times 10^{-15} \times 9.81 / 75000$ $= 6.5 \times 10^{-19} \text{ C}$ negative charge	C1 C1 A1 A1	[4]
	(iv) $F > mg$ or F now greater drop will move <u>upwards</u>	B1 B1	[2]

5	(a) (i) $I_1 + I_2 = I$	A1	[1]
---	-------------------------	----	-----